Chapter 5: Enterprise Metamodel

Study Guide, based on Mastering the TOGAF Standard by Eric Jager (2025), Chapter 5 (pp. 63-110)

Overview

Chapter 5 (pp. 63-110) explains the **Enterprise Metamodel**: the set of entities and relationships in which every artifact of the Content Framework is rooted (p. 63). Section 5.1 describes the metamodel's five sections and all 36 of its entities: 8 General Entities (5.1.1), 19 Business Architecture entities (5.1.2), and 3 entities each for Data (5.1.3), Application (5.1.4), and Technology (5.1.5) (pp. 63-101). It illustrates them with a running example of a fictional healthcare organization. Section 5.2 shows how the metamodel, the Content Framework, and the ADM are "inextricably linked", and gives five steps for tailoring the metamodel in the Preliminary Phase (pp. 101-107). Section 5.3 summarizes (pp. 107-110). Leave with this idea: the TOGAF metamodel is a *Foundation* Enterprise Metamodel, a starting point that must be tailored before you build the Content Framework (pp. 63, 106). Chapter 5 is part of the GREEN, BLUE, and GOLD reading tours (p. 63).

> **Reading note:** Entity definitions in Tables 5-1 to 5-5 come from the Standard. The longer explanations, the healthcare/HR recruitment examples, and two points are Jager's own: his strong suggestion to add a **stakeholder** entity (p. 104), and his observation that the **business information** entity may signal the return of an Information Architecture domain (p. 87). Easy to confuse: **capability** (general entity: an ability) vs. **business capability** (an ability *with a purpose*); **value stream** vs. **process**; **function** vs. **process**; **actor** vs. **organization unit**.

Detailed Section Breakdown

Chapter Opener (p. 63)

- **Root of all content.** Every entity in the (Extended) Content Framework is rooted in the Enterprise Metamodel, making it fundamental to the Standard (p. 63).
- **Foundation, not final.** No two organizations are alike, so the metamodel is a Foundation Enterprise Metamodel to be tailored, i.e., a starting point (p. 63).
- **Sections, not layers.** The Content Framework categorizes the metamodel into five interrelated sections. This is not an architecture diagram of domains or layers (p. 63).

5.1 Sections of the Enterprise Metamodel (pp. 63-65)

- **What it contains.** The most common entities and relationships likely to occur in most organizations. It gives context to the artifacts named in the ADM phase descriptions (pp. 63-64).
- **Kept in sync with the Content Framework.** The metamodel supplies the entities, the Content Framework defines work products from them, and the Content Framework evolves the metamodel (p. 64).
- **Five sections.** General Entities, Business Architecture, Data Architecture, Application Architecture, Technology Architecture (p. 64, Figure 5-1 on p. 65).

5.1.1 General Entities (pp. 65-73)

Eight entities so generic that they can be related to any domain (pp. 65-67):

Entity	Definition (Table 5-1)	Key explanation and example
Principle	Qualitative statement of intent the architecture should meet, with a name, rationale, and implications	Like a country's constitution: high-level, strategic, binding, rarely changed, based on business goals. Architecture and Enterprise Principles (Ch. 14.2) (p. 68)
Constraint	An (mostly external) factor preventing particular approaches to goals	Example: a staffing agency's standard job profiles prevent creating a new role (p. 68)
Requirement	Quantitative statement of a business need that a particular architecture or work package must meet	Specifies the objective. Guides design, development, and validation (p. 68)
Assumption	Statement of probable fact not yet fully validated, due to external constraints	Lets decisions proceed despite uncertainty, e.g., assuming an existing app supports certain requirements (pp. 69-70)
Location	A place where activities occur. Can be composed and decomposed	Physical sites or regions. Also data centres in IT distribution diagrams (p. 70)
Gap	Statement of difference between two states	Result of gap analysis in Phases B, C, D. Input to migration planning (p. 70, Figure 5-4)
Work package	Set of actions to achieve business objectives. Can be part of a project, a project, or a program	Also called an *initiative*. Closes gaps from Baseline through Transition to Target (pp. 71-72)
Capability	An ability that an organization, person, or system possesses	Central to capability-based planning: what an organization *can do*, not what it *has*. Airplane (flies), motorcycle (weaves through traffic) (p. 72)

- **Principle-requirement-constraint relationship (Figure 5-3).** Two principles frame an objective. One is detailed by a requirement and implemented by an initiative. The other is limited by a constraint, and an initiative is linked to the constraint to try to remove it (p. 69).
- **Gap example (Figure 5-4).** Application A lacks needed functionality. Application B is introduced alongside it, data and interfaces migrate, then A is decommissioned. This illustrates gaps across Baseline, Transition, and Target (p. 71).
- **Capability vs. business capability.** A business capability is an ability "to achieve a specific purpose or outcome". The added purpose makes it actionable and more meaningful, because "simply having an ability says nothing about its usefulness" (p. 73).
- **Target is not definitive.** The "desired future state" is a state at a point in time, not a final state (p. 70, footnote).

5.1.2 Business Architecture (pp. 74-92)

The largest section: 19 entities (p. 74, Table 5-2 on pp. 74-76).

Entity	Definition (abridged)	Explanation / example	Page
Driver	External or internal condition motivating the organization to define goals	Market, legislation, tech, leadership changes. Stakeholder-related drivers are called *concerns*	p. 77
Goal	High-level statement of intent, used to measure success	Qualitative: "increase", "improve"	p. 77
Objective	Aim declared in a SMART way	Details a goal, e.g., "comply with legislation by Q3 2028"	p. 77
Measure	Indicator tracked to determine success or alignment	Three types: KPIs, metrics, target values. Example: the Q3 Legal Department Report	p. 78

Entity	Definition (abridged)	Explanation / example	Page
Course of action	Direction from strategic goals and objectives, often to deliver the value proposition	What the organization *decided to do*. Strategies (long-term, broad) vs. tactics (short-term, narrow)	p. 79
Business capability	Ability a business may possess or exchange to achieve a purpose	See 5.1.1	p. 80
Value stream	End-to-end activities creating an overall result for a customer, stakeholder, or end user	Made of stages and supported by business capabilities. Example: Acquire Talent	pp. 80-83
Function	Business behaviours grouped by chosen criteria, close-coupled to org units	Logical grouping of activities. *What* is needed, broader than process	p. 83
Process	Sequence of activities achieving an outcome, decomposable	*How* things get done step by step	pp. 83-84
Organization unit	Self-contained unit of resources with goals, objectives, measures	Like a department. With goals and objectives it behaves like a *stakeholder*	pp. 84-85
Actor	Person, organization, or system with a role that initiates or interacts with activities	Can be a whole organization but not a department (that's an org unit)	p. 85
Role	Usual or expected behaviour of an actor. An actor may have several roles	Example: two HR staff with the role HR Business Partner	pp. 85-86
Product	Outcome offered to customers (materials and/or services)	Example: an employment agreement	p. 86
Contract	Agreement between consumer and provider setting functional and non-functional parameters	Formalizes roles, responsibilities, and deliverables	p. 87
Business information	A concept and its semantics used within the business	Links to BIZBOK's *information concept*. Realized by the data entity	pp. 87-89
Control	Decision step with logic to set a process's execution approach or ensure compliance	Used for governance, compliance, monitoring, risk, and change management	pp. 89-90
Event	Organizational state change that triggers processing	Example: "candidate selected" triggers Employee Hiring	p. 90
Service quality	Non-functional requirements assigned to a business, application, or technology service	Reliability, responsiveness, satisfaction. Tied to IT service management	p. 91
Business service	Encapsulates a unique element of business behaviour	The link between Business and the Data/Application/Technology architectures	p. 91

- **Driver to course of action chain (Figure 5-6).** Driver: new legislation. Goal: implement measures. Objective: by Q3. Measure: legal department report. Course of action: acquire staff (pp. 79-80).
- **Value stream vs. business process.** A value stream reflects the *business model and value proposition*: overall value-creating behaviour, not time-ordered tasks. A process reflects the *operating model*: a time-ordered sequence for an individual case, with decision points (pp. 80-81).
- **Acquire Talent example (Figures 5-7 to 5-9).** Stages: Sourcing and Attraction, Selection and Assessment, Offer and Onboarding. Each is supported by process steps from the parent processes Acquiring Talent, Evaluating Candidates, and Employee Hiring (pp. 81-84).
- **Value is intangible.** The value stream delivers *hired talent*, which is intangible. The process delivers a signed employee agreement, which is tangible. "Value streams never deliver a tangible end product. They deliver value", and value is "in the eye of the beholder" (p. 82).
- **Function vs. process (Figure 5-10).** Functions are broader and more abstract and contain processes. They are tied to activities, not departments (pp. 83-84).

- **Business service as hinge.** It sits between the strategic and tactical architectures and the operational ones (p. 91).

Jager's commentary (author's view): The business information entity's link to BIZBOK's information concept suggests a possible return of the Information Architecture domain, which disappeared after TOGAF 8 (p. 87). From a TOGAF perspective, an organization unit related to goals, objectives, and measures is best represented as a stakeholder (pp. 84-85).

5.1.3 Data Architecture (pp. 92-95)

Entity	Definition	Explanation
Data entity	Data the business recognizes as a distinct concept	Modelled in ERDs or class diagrams. Accessed by a business service, consumed by an actor, and realizes business information (p. 93)
Logical data component	Data structure of logically related data entities	Technology-agnostic: *what* the data is (p. 94)
Physical data component	Realizes logical data components in a technology's format or schema	Databases, warehouses, cloud storage (p. 95)

- **Data to information (Figure 5-15).** The numbers 22, 23, 22, 25, 21 become information only with context ("next five days' temperatures"). Linking a data entity to business information turns data into value (p. 94).
- **Equivalence.** According to the Standard, the data entity corresponds to BIZBOK's information concept (p. 89).

5.1.4 Application Architecture (pp. 96-98)

Entity	Definition	Explanation
Application service	The automated elements of a business service	Delivers or supports all or part of one or more business services (p. 97)
Logical application component	Application functionality defined by services offered and data maintained, independent of technology	E.g., "HR recruitment system" without choosing a product (pp. 97-98)
Physical application component	Realization of logical functionality via applications that may be hired, procured, or built	Installed software, configuration, deployment (p. 98)

- **Relationship (Figure 5-18).** The logical component implements the application service, which supports a business service and may use data entities. The product choice happens in physical design (pp. 97-98).

5.1.5 Technology Architecture (pp. 98-101)

Entity	Definition	Example
Technology service	Technical capability providing infrastructure that supports applications	Cloud storage service (p. 99)
Logical technology component	Implementation-independent encapsulation of technology services	"DBMS" without naming a product (p. 100)
Physical technology component	Realization using a particular deployable technology product	MySQL realizing the DBMS (pp. 100-101)

- **Logical-to-physical pattern.** The same pattern recurs in Data, Application, and Technology: service, then logical component (what), then physical component (which product) (pp. 94-101).

5.2 Tailoring the Enterprise Metamodel (pp. 101-107)

- **The triangle (Figure 5-21).** ADM, Enterprise Metamodel, and Content Framework are inextricably linked. The ADM (Preliminary Phase) creates the organization-specific metamodel. That metamodel feeds the Content Framework. The Content Framework's deliverables are then produced in the ADM phases (pp. 101-105).
- **Add and remove.** Architects may drop irrelevant entities and relationships or add new ones. The Standard "does not aim to constrain an enterprise's selection of artifacts" (p. 103).
- **Timing.** This happens in the Preliminary Phase, the first ADM iteration, while the EA Capability is established. Many architects overlook this step (pp. 103-104).
- **Suggested additions.** Stakeholder, business collaboration, business interaction, business interface, resource, risk, and value. Stakeholder is used in the Stakeholder Catalog, and most of the others appear in the Business Model Diagram (Ch. 12.4.2) (p. 105).

Step	What to do	Page
1. Understand the Enterprise Metamodel	Learn the Foundation Enterprise Metamodel's standard entities and relationships	p. 106
2. Customize the metamodel	Add exceptional entities unique to the organization	p. 106
3. Integrate the customized metamodel	Use it as input to the Content Framework, adding or removing deliverables	p. 106
4. Align deliverables	Produce ADM deliverables according to the customized Content Framework	p. 106
5. Continuously evolve	Regularly review and update both the metamodel and the Content Framework	p. 106

Jager's commentary (author's view): He "strongly" suggests adding a **stakeholder** entity. The Standard implicitly uses the organization unit for this, while ArchiMate has an explicit stakeholder element, and adding it improves TOGAF-ArchiMate integration (p. 104). "Tailoring the Enterprise Metamodel is an absolute necessity" (p. 106).

5.3 Summary (pp. 107-110)

- Five sections, 36 entities: 8 general, 19 business, and 3 each for data, application, and technology (pp. 107-108).
- **TOGAF vs. ArchiMate.** ArchiMate classes driver, goal, objective, and measure as *motivation elements*. TOGAF puts them all under Business Architecture because it has no separate motivation or strategy layer, only the four domains (p. 108).
- The metamodel is input to both the Content Framework and the ADM, and it takes time and care to tailor (p. 110).

Key Terms & Definitions

Term	Definition	Page
Enterprise Metamodel	Model of the common entities and relationships in which all architecture content is rooted	pp. 63-64
Foundation Enterprise Metamodel	TOGAF's standard starting set of 36 entities, to be tailored	pp. 63, 108
Entity	An architecture concept with its own definition, belonging to one metamodel section	p. 65

Term	Definition	Page
General Entities	Eight cross-domain entities: principle, constraint, requirement, assumption, location, gap, work package, capability	pp. 65-67
Business capability	An ability a business may possess or exchange to achieve a specific purpose or outcome	p. 73
Value stream	End-to-end activities creating an overall result for a customer, stakeholder, or end user, made of stages	p. 80
Concern	A stakeholder-related driver	p. 77
SMART	Specific, Measurable, Achievable, Realistic, Timebound	p. 77
Business service	Entity linking the Business Architecture to the Data, Application, and Technology Architectures	p. 91
Information concept (BIZBOK)	High-level abstraction of information the business uses, which corresponds to the TOGAF data entity	pp. 88-89
Logical vs. physical component	Technology-independent description of *what* vs. the concrete realization by a product	pp. 94-101

Figures & Tables Referenced

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Figure 5-21	Relation between ADM, Enterprise Metamodel, and Content Framework	p. 102

Self-Check Questions

Q: Name the five sections of the Enterprise Metamodel and how many entities each has.

A: General Entities (8), Business Architecture (19), Data (3), Application (3), Technology (3), for 36 in total (pp. 64, 107-108).

Q: How does a business capability differ from a capability?

A: A business capability is an ability "to achieve a specific purpose or outcome". The purpose makes it actionable and valuable, whereas a plain capability is just an ability (p. 73).

Q: How does a value stream differ from a business process?

A: A value stream shows overall value creation from the stakeholder's perspective and reflects the business model. It is not time-ordered and delivers intangible value. A process is a time-ordered sequence for an individual case, reflects the operating model, and can produce tangible outputs (pp. 80-82).

Q: Trace the Figure 5-6 example from driver to course of action.

A: Driver: new legislation. Goal: implement measures. Objective: done by Q3. Measure: a Q3 Legal Department Report. Course of action: acquire the staff needed (pp. 79-80).

Q: Which entity connects the Business Architecture to the other three domains?

A: The business service (pp. 91-92).

Q: What are the three types of measures?

A: KPIs (outcome-focused, tied to CSFs), metrics (broader, operational), and target values (desired performance levels) (p. 78).

Q: When and where is the organization-specific metamodel created?

A: In the Preliminary Phase, during the first ADM iteration, while the Architecture Capability is established (pp. 101, 103-104).

Q: What entities does Jager suggest adding to the metamodel, and why stakeholder in particular?

A: Stakeholder, business collaboration, business interaction, business interface, resource, risk, and value. Stakeholder is used in the Stakeholder Catalog, the Standard only covers it implicitly through organization unit, and adding it improves alignment with ArchiMate (pp. 104-105).

Q: Why does TOGAF put driver, goal, objective, and measure under Business Architecture when ArchiMate calls them motivation elements?

A: TOGAF has no separate motivational or strategic elements. It organizes everything by the four architecture domains (p. 108).

Q: List the five steps for tailoring the Enterprise Metamodel.

A: Understand the metamodel; customize it; integrate it into the Content Framework; align deliverables during the ADM; continuously evolve (p. 106).

Q: In the DBMS example, what are the logical and physical technology components?

A: Logical: "Database Management System", with no product named. Physical: MySQL, the product that realizes it (p. 100).

Q: What is the difference between an actor and an organization unit?

A: An actor can represent a person, system, or an entire organization, but not a department within it. Departments are organization units (p. 85).